

The Lipschitz Approximation Properties in arXiv:2512.06317 Are Automatic

Automated discovery packet

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Source question

The final open-question list of arXiv:2512.06317 asks for examples of Banach spaces without the Lipschitz approximation property or without the Lipschitz p -approximation property, and asks what can be concluded about the reverse implications in the displayed implication diagram.

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- We observed the facts that $X^\#$ or $F(X)$ has approximation property implies X has $Lip - a.p$ and also that $F(X)$ has $Lip - a.p$ implies X has $Lip - p.a.p$. These would be interesting questions to analyze the converse part of those facts.
- Examples of the Banach spaces without having $Lip - a.p$, $Lip - p.a.p$.
- We have seen that $a.p \implies Lip - a.p \implies Lip - p.a.p$; also, $a.p \implies p.a.p \implies Lip - p.a.p$. So the immediate question to ask is, What can we conclude about the other-way implications for all the cases?

We leave these questions open for the time being.

Result

The question has a simple obstruction: with the definitions used in the paper, the Lipschitz approximation property has no content unless one also imposes a uniform bound on the Lipschitz constants of the approximating maps.

Theorem 1. *Let $(M, 0)$ be a pointed metric space, let X be a Banach space, and let $f \in \text{Lip}_0(M, X)$. For every compact set $C \subset M$ and every $\varepsilon > 0$, there is a map $g \in \text{Lip}_{0\mathcal{F}}(M, X)$ such that*

$$\sup_{x \in C} \|g(x) - f(x)\| < \varepsilon.$$

Consequently, every Banach space has the Lipschitz approximation property. Since every relatively p -compact set is compact, every Banach space also has the Lipschitz p -approximation property, for every $1 \leq p < \infty$.

Proof. We use a finite-dimensional Lipschitz approximation of the identity on compact subsets of the range. Let $K \subset X$ be compact and contain 0. We first show that, for every $\varepsilon > 0$, there are a finite-dimensional subspace $E \subset X$ and a Lipschitz map $R : K \rightarrow E$ such that $R(0) = 0$ and

$$\sup_{y \in K} \|R(y) - y\| < \varepsilon.$$

Put $\delta = \varepsilon/16$ and $A = \{y \in K : \|y\| \geq \delta\}$. If A is empty, take $R = 0$. Otherwise choose a finite $\delta/2$ -net $a_1, \dots, a_N$ in A and set $E = \text{span}\{a_1, \dots, a_N\}$. For $y \in A$ define

$$\theta_i(y) = (\delta - \|y - a_i\|)_+, \quad D(y) = \sum_{i=1}^N \theta_i(y),$$

and

$$q(y) = D(y)^{-1} \sum_{i=1}^N \theta_i(y) a_i.$$

The net condition gives $D(y) \geq \delta/2$ on A , so q is Lipschitz on A . Moreover, whenever $\theta_i(y) > 0$ we have $\|a_i - y\| < \delta$, hence

$$\|q(y) - y\| \leq \delta \quad (y \in A).$$

Let

$$\lambda(y) = \min\{1, \max\{0, (\|y\| - \delta)/\delta\}\}$$

and define $R(y) = 0$ if $\|y\| \leq \delta$, while $R(y) = \lambda(y)q(y)$ on A . This is Lipschitz on K . Indeed, on A it is a product and quotient of Lipschitz functions with denominator bounded below. If $x \in A$ and $\|y\| \leq \delta$, then

$$\|R(x) - R(y)\| = \lambda(x)\|q(x)\| \leq \frac{(\|x\| - \delta)_+}{\delta}(\|x\| + \delta) \leq \frac{M + \delta}{\delta}\|x - y\|,$$

where $M = \sup_{z \in K} \|z\|$; the last inequality follows from $\|x - y\| \geq \|x\| - \|y\| \geq \|x\| - \delta$. Thus the two Lipschitz pieces glue with a finite constant. The approximation estimate is immediate. If $\|y\| \leq \delta$, then $\|R(y) - y\| \leq \delta$. If $\delta < \|y\| < 2\delta$, then

$$\|R(y) - y\| \leq \|q(y)\| + \|y\| \leq (\|y\| + \delta) + \|y\| < 5\delta.$$

If $\|y\| \geq 2\delta$, then $R(y) = q(y)$ and $\|R(y) - y\| \leq \delta$. In all cases the error is below ε .

Because E is finite-dimensional, the Lipschitz map $R : K \rightarrow E$ extends to a Lipschitz map $\tilde{R} : X \rightarrow E$ with $\tilde{R}(0) = 0$: choose a basis of E , extend the finitely many scalar coordinate functions by the McShane extension theorem, and recombine them.

Now apply this construction to the compact set $K = f(C) \cup \{0\} \subset X$. The map

$$g = \tilde{R} \circ f$$

is Lipschitz, vanishes at 0, and has range contained in the finite-dimensional space E , so $g \in \text{Lip}_{0\mathcal{F}}(M, X)$. For $x \in C$,

$$\|g(x) - f(x)\| = \|\tilde{R}(f(x)) - f(x)\| < \varepsilon.$$

Taking $M = X$ and $f = \text{Id}_X$ proves the Lipschitz approximation property for every Banach space. The p -version follows because the topology τ_p tests approximation only on relatively p -compact sets, and these sets are compact. $\square$

Corollary 1. *There are no Banach spaces without the Lipschitz approximation property, and there are no Banach spaces without the Lipschitz p -approximation property, with the unbounded definitions used in arXiv:2512.06317.*

Corollary 2. *The reverse implications*

$$\text{Lip -a.p.} \implies \text{a.p.} \quad \text{and} \quad \text{Lip -}p\text{-a.p.} \implies \text{a.p.}$$

are false. Indeed, let E_0 be Enflo's Banach space without the classical approximation property. By the theorem, E_0 has both Lipschitz approximation properties, but it does not have the classical approximation property.

For $1 \leq p \leq 2$, the reverse implication

$$p\text{-a.p.} \implies \text{a.p.}$$

is also false. In this range, the paper cites the Karn–Sinha theorem that every Banach space has p -a.p.. Hence Enflo's space again supplies a counterexample.

Verification notes

The proof is deterministic. The helper script `code/check_partition_construction.py` mirrors the finite partition construction on a compact sample in $\mathbb{R}^2$ and checks the advertised error bound numerically. The script `code/make_open_problem_crop.py` recreates the source screenshot from `source_paper.pdf`.

References

- [1] A. Mandal, *Approximation property in terms of Lipschitz maps via tensor product approach*, arXiv:2512.06317, 2025.
- [2] P. Enflo, *A counterexample to the approximation problem in Banach spaces*, Acta Mathematica 130 (1973), 309–317.
- [3] D. P. Sinha and A. K. Karn, *Compact operators whose adjoints factor through subspaces of ℓ_p* , Studia Mathematica 150 (2002), 17–33.